Project Proposal

Management System for Medical Centre

Department of Computing
Faculty of Science
Eastern University Sri Lanka
2023

CS2071 – Case Study Based
Software Development Project
Group 4

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1. Introduction

In the labyrinth of academia, where the pursuit of knowledge intertwines with the well-being of its community, the university medical center stands as a sentinel of health, offering a sanctuary for students, faculty, and staff. However, the noble mission of these centers is not without its complexities and challenges. As the world of healthcare evolves, so too must the systems that support it. In this light, the imperative to revolutionize the healthcare management paradigm at our university has given rise to a visionary initiative– the creation of a robust and sophisticated Medical Center Management System (MCMS). This transformative system seeks not only to address existing inefficiencies but to set new standards in healthcare management, enriching the experience of those within the academic community.

Universities are microcosms of diversity, housing individuals with distinct needs, schedules, and health requirements. The university medical center, therefore, becomes a nexus where the academic and health dimensions of life converge. Against the backdrop of this convergence, a myriad of challenges has emerged, demanding innovative solutions that go beyond the conventional paradigms of healthcare management.

A primary challenge lies in the unpredictable nature of doctor schedules. Students, faculty, and staff often find themselves grappling with uncertainties about when a particular medical professional will be available. This lack of transparency not only leads to prolonged waiting times but also hampers the overall accessibility of healthcare services within the university.

Queue management during medicine distribution poses another significant hurdle. The bottleneck created during the availability of doctors exacerbates the issue, resulting in lengthy queues and frustrated patients. This not only impacts the efficiency of the medical center but also diminishes the quality of the healthcare experience for those seeking pharmaceutical assistance.

The modern age demands a shift towards more accessible and user-friendly healthcare solutions. The traditional models of seeking medical assistance, often necessitating physical visits to the medical center, are increasingly incongruent with the expectations of a digitally literate university community. The need for a seamless and accessible online medical management system has become paramount.

2. Background and Motivation

Within the dynamic intersection of academia and healthcare, the university medical center emerges as a vital nexus for the health and well-being of students, faculty, and staff. The journey of these medical centers, from humble infirmaries to comprehensive healthcare hubs, has been marked by evolution and adaptation to the changing needs of the university community. However, the current healthcare management practices face challenges in meeting the expectations of a digitally literate and diverse community.

In the twenty-first century, heightened expectations and a paradigm shift in healthcare demands necessitate a reevaluation of existing systems. The conventional models, designed for basic medical needs, are encountering the complexities of modern healthcare expectations. Challenges such as unpredictable doctor schedules, queues for medicine distribution, limited accessibility of medical services, and gaps in epidemic preparedness underscore the need for innovative and technology-driven solutions.

The unpredictable nature of doctor schedules results in uncertainties among patients, leading to prolonged waiting times and diminished healthcare experiences. The distribution of medicine during doctor availability often leads to lengthy queues and congestion, hindering efficient pharmaceutical assistance. Accessibility of medical services, reliant on physical visits to the medical center, falls short in meeting the expectations of a tech-savvy community. Epidemic preparedness poses unique challenges, requiring swift and informed decision-making and efficient management of health information, particularly for vulnerable populations.

The motivation to embark on the transformative journey towards the development of a comprehensive Medical Center Management System (MCMS) is rooted in a commitment to excellence and a recognition that technology and innovation can elevate healthcare management to new heights. The goal is to empower every member of the university community with a healthcare experience that anticipates and exceeds expectations. The MCMS is conceived as a solution that not only resolves existing challenges but serves as a beacon of progress, attuned to the evolving needs and preferences of the university community.

3. Problem Specification

The current state of the university medical center reveals a series of challenges that impact the accessibility, efficiency, and overall quality of healthcare services. Notably, the lack of transparency in doctor schedules leads to confusion and extended waiting times for students seeking medical attention. The distribution of medicine during doctor availability exacerbates the issue, causing congestion and delays. Additionally, there is a need to enhance the accessibility of medical services, especially during epidemic cases, and to streamline the process of obtaining health reports for children. These challenges collectively underscore the necessity for a comprehensive solution that addresses these pain points and transforms the medical management landscape.

4. Aim and Objectives

4.1 Aim

The overarching aim of this project is to develop and implement a robust Medical Center Management System (MCMS) for our university. This system aspires to redefine the healthcare management paradigm, fostering transparency, efficiency, and accessibility. By integrating real-time data, user-friendly interfaces, and advanced functionalities, the aim is to create a model healthcare system that not only addresses existing challenges but sets a new standard for medical management within the university community.

4.2 Objectives

- Real-Time Doctor Availability:
  - Develop a dynamic scheduling system that enables doctors to update their availability in real time.
  - Implement a user-friendly interface for students to view and schedule appointments based on the real-time availability of doctors.

- Online Appointment Scheduling:
  - Introduce an online appointment scheduling system that minimizes waiting times and aligns with modern preferences for digital interactions.
  - Enable seamless appointment bookings, cancellations, and rescheduling through a user-centric platform.

- Queue Management for Medicine Distribution:
    - Incorporate a queue management system for the efficient distribution of medicine during doctor availability.
    - Minimize waiting times by optimizing the process of allocating and managing queues through technological solutions.

- Online Medical Management:
    - Develop a comprehensive online medical management module, allowing students to seek medical assistance, input symptoms, and receive guidance or prescriptions remotely.
    - Ensure a secure and user-friendly interface for seamless interaction with medical professionals.

- Health Records Management System:
    - Enable secure access to electronic health records for parents or guardians, particularly during epidemic cases.
    - Facilitate collaboration between medical professionals and families by providing a dedicated portal for accessing crucial health information.

- Data Analytics for Epidemic Cases:
    - Implement cutting-edge data analytics tools to monitor and analyze health trends during epidemic cases.
    - Contribute to proactive decision-making, resource allocation, and community health management through data-driven insights.
    - Through the achievement of these objectives, the project aims to deliver a Medical Center Management System that not only resolves existing challenges but also empowers our university community with a modern, efficient, and patient-centric healthcare management solution.

5. Proposed Solution

5.1 Technology Adapted

- MongoDB (Database): MongoDB is chosen for its flexibility in handling unstructured data, scalability, and seamless integration with Node.js. Its JSON-like document structure aligns well with the complex data structures present in healthcare management.

- Express.js (Backend Framework): Express.js serves as the backend framework, providing a minimalist and flexible architecture for building robust APIs. Its simplicity streamlines the development of RESTful APIs and middleware, ensuring efficient communication between the frontend and the database.

- React (Frontend Library): React is employed for its declarative and efficient approach to building user interfaces. Its component-based architecture promotes modularity, reusability, and efficient rendering through the use of a virtual DOM.

- Node.js (Runtime Environment): Node.js facilitates server-side JavaScript execution, maintaining a consistent language throughout the entire application stack. Its non-blocking, event-driven architecture enhances performance and handles concurrent requests effectively.

- Material-UI (MUI): Material-UI is chosen for UI development due to its adherence to Google's Material Design principles, React component-based structure, and the flexibility it offers for theming and customization. It ensures a consistent and visually appealing design language for the entire application.

5.2 Nature of Solution

Expected Input:

- User Data: Information about students, faculty, staff, and medical professionals.
- Doctor Schedules: Real-time availability of doctors.
- Medical Records: Electronic health records and historical data.
- Appointment Requests: User-generated requests for medical appointments.
- Symptoms and Requests: Input from users regarding medical symptoms and requests for assistance.

Process:

- Real-Time Data Updates: Doctors update their availability in real time.
- Online Appointment Scheduling: Users schedule appointments through an intuitive interface.
- Queue Management: Efficient distribution of medicine during doctor availability.
- Online Medical Management: Users input symptoms and receive guidance or prescriptions remotely.
- Electronic Health Records (EHR) Access: Secure access to health records for parents or guardians.
- Data Analytics: Monitoring and analyzing health trends, especially during epidemic cases.

Expected Output:

- Real-Time Doctor Availability: Users can view and schedule appointments based on updated doctor availability.
- Online Appointment Confirmations: Automated confirmations and reminders for scheduled appointments.
- Queue Management: Optimized distribution process, minimizing waiting times.
- Online Medical Assistance: Remote medical guidance and prescriptions.
- EHR Access: Secure and seamless access to electronic health records.
- Data Analytics Insights: Informed decision-making during epidemic cases.

Users:

- Students, Faculty, and Staff: Utilize the system for scheduling appointments, seeking medical assistance, and accessing health records.
- Medical Professionals: Update real-time availability, manage appointments, and provide remote medical assistance.
- Parents or Guardians: Access health records and information about their dependents.

5.3 Feasibility of Implementation

The implementation of the proposed Medical Center Management System (MCMS) exhibits strong feasibility on various fronts. From a technical perspective, the adoption of the MERN stack and Material-UI ensures a streamlined development process. The cohesive integration of MongoDB for data storage, Express.js for backend development, React for frontend interfaces, and Node.js for runtime execution creates a harmonious and scalable architecture. Material-UI, with its React component-based structure, provides an efficient and visually appealing user interface. The extensive documentation and active community support for these technologies enhance their feasibility.
Operationally, the feasibility is bolstered by a comprehensive training and onboarding program, ensuring that medical professionals, administrative staff, and users can navigate the system effectively. The commitment to regular maintenance and updates aligns with the evolving needs of the healthcare landscape and emerging technological advancements.

From an economic standpoint, the open-source nature of many components in the chosen stack minimizes licensing costs. Furthermore, the envisioned efficiency gains, reduction in waiting times, and improved healthcare management contribute to long-term cost-effectiveness.
In terms of legal and ethical considerations, the proposed system adheres rigorously to data protection regulations. Robust security measures will be implemented to safeguard sensitive health information, ensuring compliance with legal requirements and ethical standards.

In summation, the proposed MCMS stands as a feasible solution that combines technological sophistication with operational, economic, and ethical viability. The careful selection of technologies and a strategic approach to implementation position the system to not only meet the current challenges faced by the university medical center but also to adapt and thrive in the evolving landscape of healthcare management.

6. Resource Requirement

- Internet Connectivity
- Computer with minimum 4GB RAM and 3.0GHz processing power
- Domain Name
- Hosting Service
- MongoDB Server

7. Reference

[1] Health Center Eastern University Sri Lanka: https://www.esn.ac.lk/life-at-eusl/health-centre

[2] MongoDB: The database for giant ideas. https://www.mongodb.com/

[3] Express.js: Fast, unopinionated, minimalist web framework for Node.js. https://expressjs.com/

[4] React: A JavaScript library for building user interfaces. https://reactjs.org/

[5] Node.js: https://nodejs.org/

[6] Material-UI: React components for faster and simpler web development. https://material-ui.com/

8. Appendix

Task	November							
	23	24	25	26	27	28	29	30
Requirement analyzing and planning								
Database Design								

Task	December														
	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15
Database Design															
Backend Development															
UI/UX Development															

Task	December														
	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Backend Development															
UI/UX Development															
Frontend Development															

Task	January														
	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15
Frontend Development															
Testing and Bug fixing															

9. Assignment of Tasks

Index Number	Name	Tasks
PS 2826	B.W.J.N. Bandara	• Requirement analyzing and planning • Database Design • Backend Development • Frontend Development – Appointment, Online assistant system, Analysis and showing data using charts • Testing - Appointment, Online assistant system, Analysis and showing data using charts
PS 2776	H.A.K.Buddhika	• Requirement analyzing and planning • Database Design • UI/UX Development – Login • Frontend Development – Login • Testing – Login
PS 2822	K.A.I.R.Perera	• Requirement analyzing and planning • Database Design • UI/UX Development – Home • Frontend Development – Home • Testing – Home
PS 2882	S.D.S.P.Kumari	• Requirement analyzing and planning • Database Design • UI/UX Development – Profiles • Frontend Development – Profiles • Testing – Profiles
PS 2811	J.Sansitha	• Requirement analyzing and planning • Database Design • UI/UX Development – Appointments • Frontend Development – Appointments • Testing – Appointments
PS 2877	R.M.L.M. Gunawardhana	• Requirement analyzing and planning • Database Design • UI/UX Development – Health record section • Frontend Development – Health record section • Testing – Health record section

PS 2760	D.M.V.Piyumal	• Requirement analyzing and planning • Database Design • UI/UX Development – Online assistant system • Frontend Development – Online assistant system • Testing – Online assistant system
PS 2806	A.M.Afran	• Requirement analyzing and planning • Database Design • UI/UX Development – Analysis and showing data using charts • Frontend Development – Analysis and showing data using charts • Testing - Analysis and showing data using charts

10. Signatures of Group Members

Index numbers	**Name**	**Signature**
PS 2826	B.W.J.N.Bandara	
PS 2776	H.A.K.Buddhika	
PS 2822	K.A.I.R.Perera	
PS 2882	S.D.S.P.Kumari	
PS 2811	J.Sansitha	
PS 2877	R.M.L.M. Gunawardhana	
PS 2760	D.M.V.Piyumal	
PS 2806	A.M.Afran	

11. Supervisors Declaration

I hear by declare that I have checked this project and, in my opinion, this project is adequate in terms of scope and quality

1. Name of lecturer in charge: Mrs. S.Thavareesan

   Designation:

   Date:

   Signature:

   Any further comments:

2. Name of lecturer in charge: Mr. T.Vinothraj

   Designation:

   Date:

   Signature:

   Any further comments: